Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate the defect of radiator hose cut from Marazzo model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**01.04.19**	**01.04.19**
Completion Date	**30.11.19**	**30.11.19**

Current Status: **Closed**

Cross Functional Team

Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Vikrant Ghule	L7-O	STCF	9423584194
2	**Mentor**	Mandar Joshi	L6-M	STCF	9689003630
3	**Member 1**	Ganesh Bendale	L7-O	STCF	9850952219
4	**Member 2**	Suryabhan Sor	L7-O	STCF	9763003635
5	**Member 3**	Abhijit Puranik	L7-M	SQA	9890618172
6	**Member 5**	Kiran Jadhav	L7-O	PVT	8830736147

Step1. Select Subject

PROJECT TITLE :-

To eliminate the defect of radiator hose cut from Marazzo model

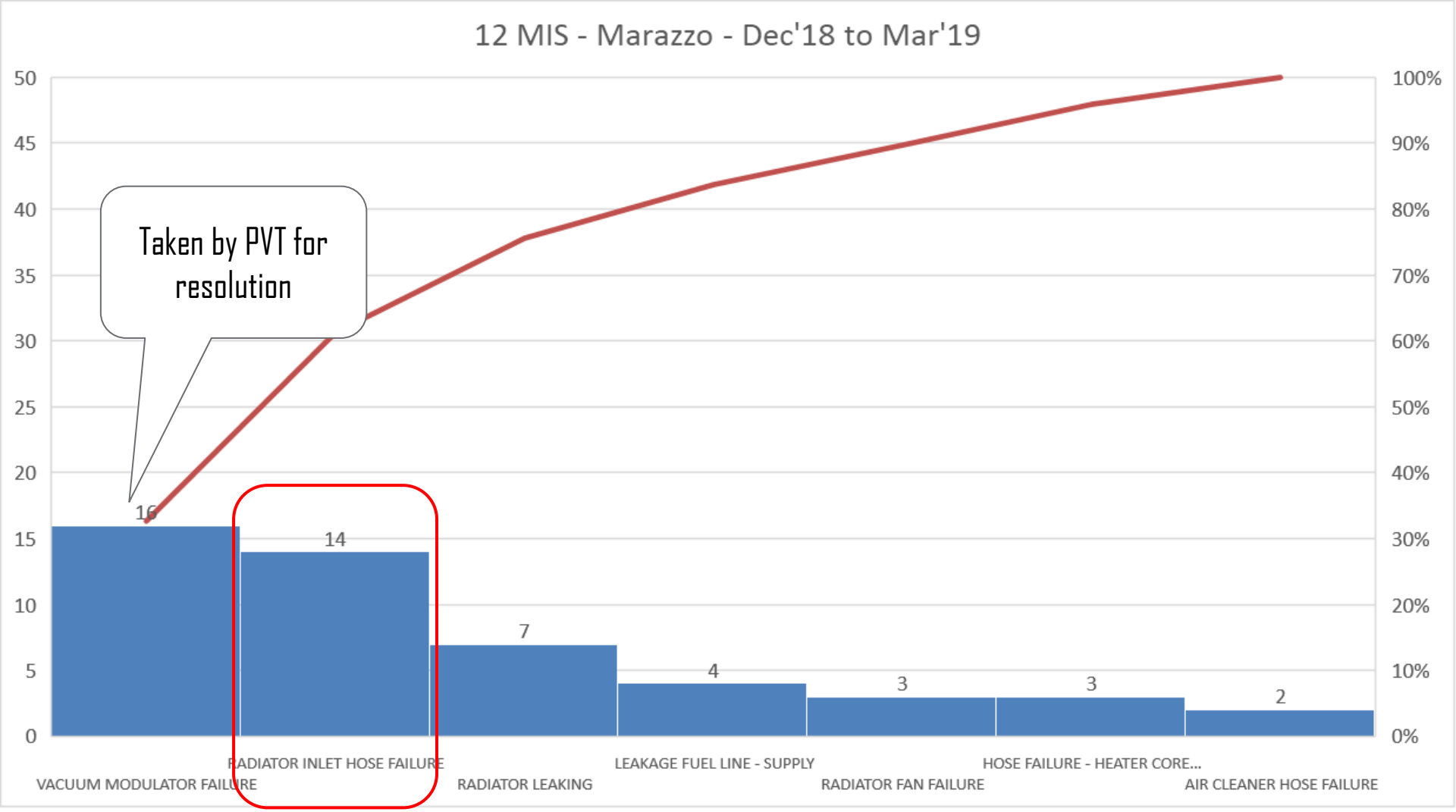
Brief Description :

- What : Radiator hose cut in Scorpio model
- Where : In field
- When : In field at high kms after fouling with battery tray
- Who :By customer
- Why : Repeat issue in field which Impact on Marazzo image & Customer dissatisfaction
- How : 14 Cases reported in field

2. Justify the choice (reasons)

Sector Objective :-
- J D Power score
Departmental Objective :-
– 3MIS & 12 MIS Improvement
- Customer satisfaction

2. Justify the choice



Inference – Radiator inlet hose failure is top issue

2. Justify the choice

Cost of Poor Quality-Rework

Severity & Consequences of Concern :-

If the radiator hose gets cut, coolant will leak. This will result in vehicle overheating and engine will cease.

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	10
Total Required Rework time per day	Min	100
Effective working time	Min	465
Operator required /day	nos	0.22
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	322.58
Avg no. of working days	days	25
Operator cost /month	Rs	8064.52
Operator Cost/year	**Rs**	**96774.19**
Total Rework cost	Rs / Year	96774

2.b Immediate Containment Action

- Awareness given to all stage , buyoff and CAI rework associates
- Check point addition at FAI, inhouse defect

Step 3. Understanding Current Situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : STCF
Cell / BWT / Line: FAI/11

Requested By- MQA | Date - 04.04.19 | Source -G nova c | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Radaitor hose cut due to fouling with battery tray

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- 19359 | Model : Marazzo

Date of Incidence/ | Shift / Time :

Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :
All WIP vehicles checked, WIP vehicle retorquing done.

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - BWT . : 9

Does a control already exist ? ☐ YES ☑ NO

Stage / Post Analysis -

Operator Name : S S Gangurde

Man(Human Action)

Is operator trained ?	☑ YES ☐ NO
Operator level (L.L.U)	I
OR Dexterity level(1…4)	L3
Is operator under training ?	☐ YES ☑ NO
Is operator temporary for Stage ?	☐ YES ☑ NO
Bad rework ?	☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ?	☑ YES ☐ NO
Last calibration dates	
Facility /Tool / Fixture in good condition ?	☑ YES ☐ NO
Is it a substitute facility ?	☐ YES ☐ NO
Are machine parameters known ? (Record in Analysis table)	☑ YES ☐ NO
Parameters are respected ?	☑ YES ☐ NO

Method(process)

Main steps are respected ?	☑ YES ☐ NO
Is there a related key point in SOP / SOS?	☑ YES ☐ NO
Are the reasons for key point known ?	☑ YES ☐ NO
Are operating reqmnts respected ?	☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ?	☑ YES ☐ NO
Was it detected by inspector/ Buoff Opr ?	☐ YES ☐ NO
In only one shift ?	☐ YES ☑ NO
In only one line ?	☑ YES ☐ NO
The problem follows the vehicle ?	☐ YES ☑ NO
The problem follows the part ?	☑ YES ☐ NO
Is it a repeat problem ?	☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table)	☑ YES ☐ NO

Product Failure

Other

Is there a Control Plan ? (Record in Analysis table)	☑ YES ☐ NO
Is the packaging standard defined for the part ?	☑ YES ☐ NO
Is the packaging standard respected ?	☑ YES ☐ NO
Is there any alert mechanism for failure ?	☐ YES ☑ NO
Buy off -	
Is the specified product used ?	☑ YES ☐ NO
Did it evolve / Change ?	☐ YES ☑ NO
Is the storage SOS respected ?	☑ YES ☐ NO
Is the shelf life / expiry date respected ?	☑ YES ☐ NO
Is the part as per FIFO?	☑ YES ☐ NO

Material/ Consumable

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

Vehicle reported for radiator hose leakage in filed. When check pint added at FAI, vehicles found with less gap/fouling between radiator hose & battery hose.

History :- No change in both parts radiator hose/battery tray from launch.

Traveller card - There was no rework related to TC hose in traveller card for all vehicles.

Detection -After air cleaner fitment this joint gets hidden. Checkpoint added at FAI console stage.

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☑ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
1. Awareness given to all stage and buyoff associates through DRF. 2. All wip vehicles checked	AT/ND	26.02.20	Closed
2. Permanent Action -	Resp	Date	Status
L1 analysis required	VRG		Open

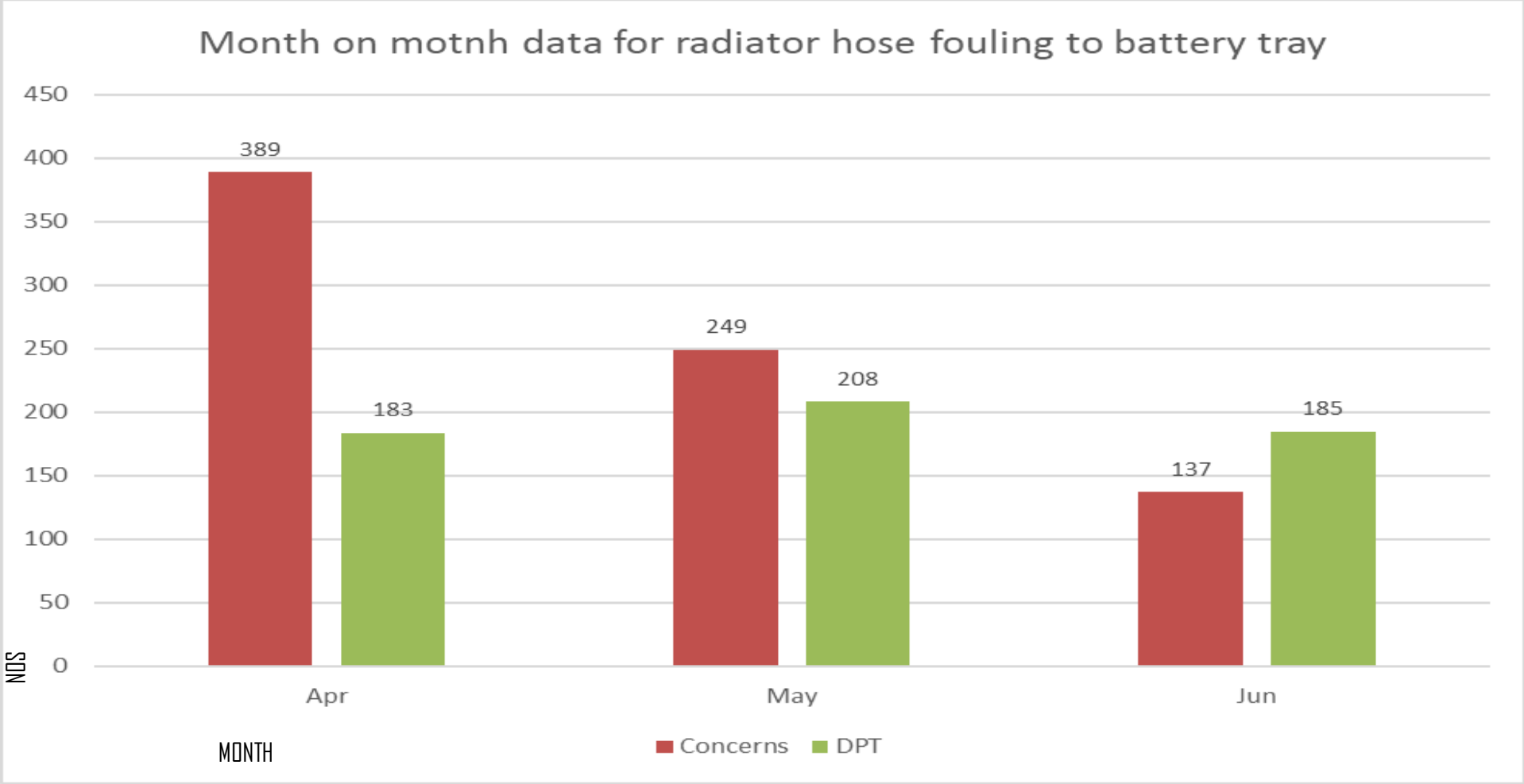
Analysed by Name/s - Vikrant Ghule
Date - 04.04.19

Choice : ☑ WSS ☐ QC Story

Inference from L0,
Problem follows part,
Problem is a repeat problem &
hence detailed analysis required

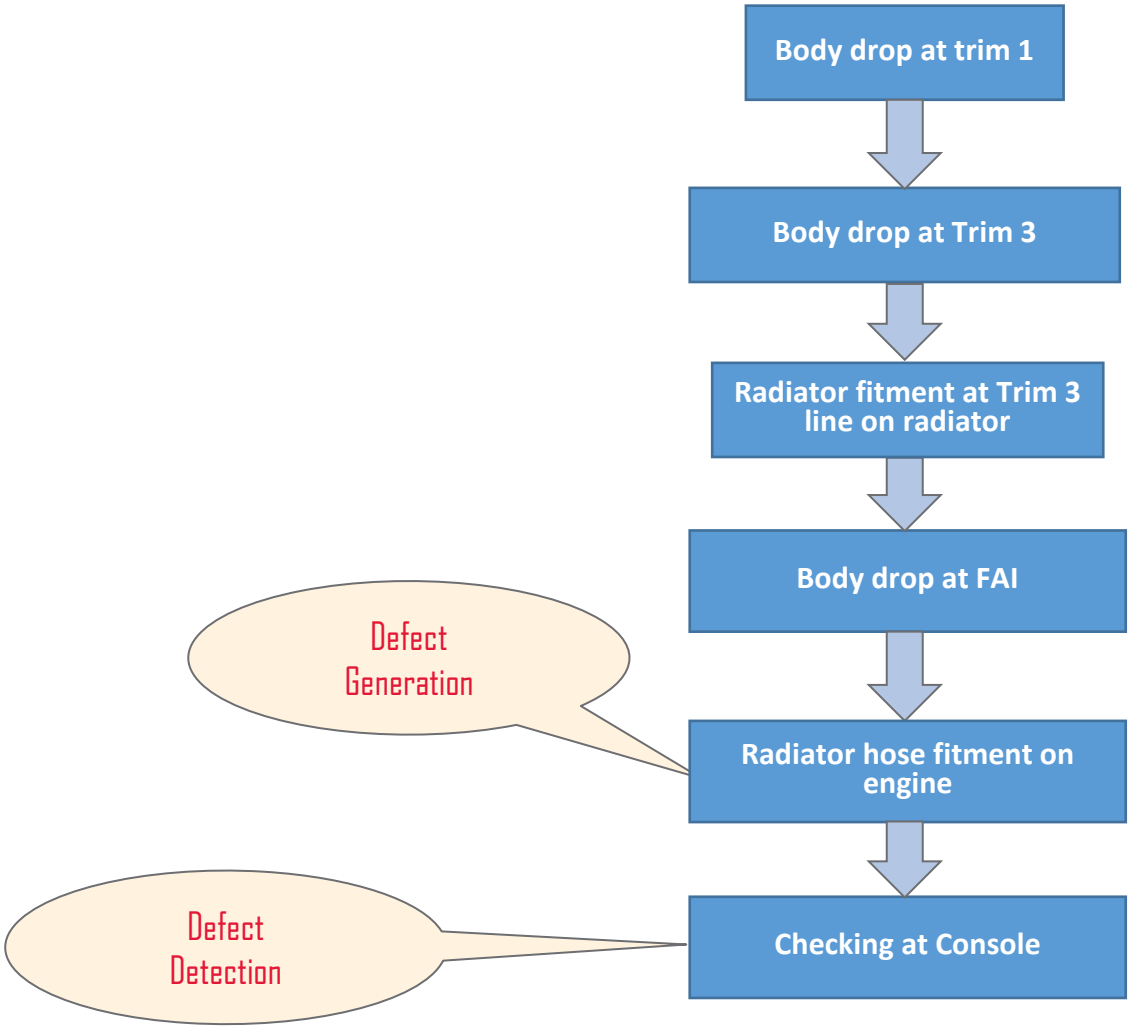
4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L4
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	**Problem follows part**
Material	Handling and shelf life of part adhered

Step 3. Understanding Current Situation



Inference – As per graph though month on month defect is reducing, but if we compare DPT , defect is constant.

Step 3. Understanding Current Situation



Step 3. Understanding Current Situation

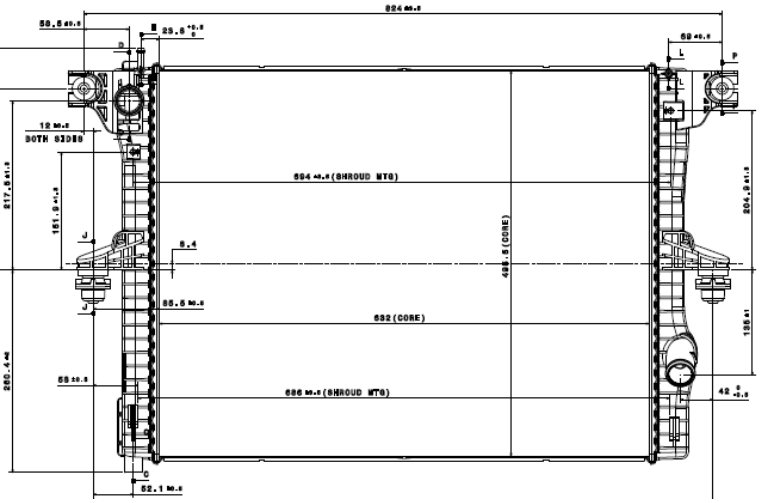
Good & Not so good observations

- Issue observed only in Marazzo only
- Associate is dexterity L4
- Associate follows main step and key points
- Associate is following his SOS while doing fitment of radiator hose
- Part storage found according to std.
- Part not damaged in storage
- Concern observed in both shifts

Step 3. Understanding Current Situation

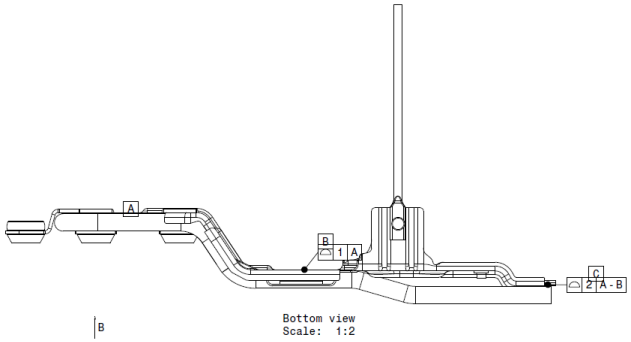
Parts Involved in the analysis:

Radiator



Radiator hose

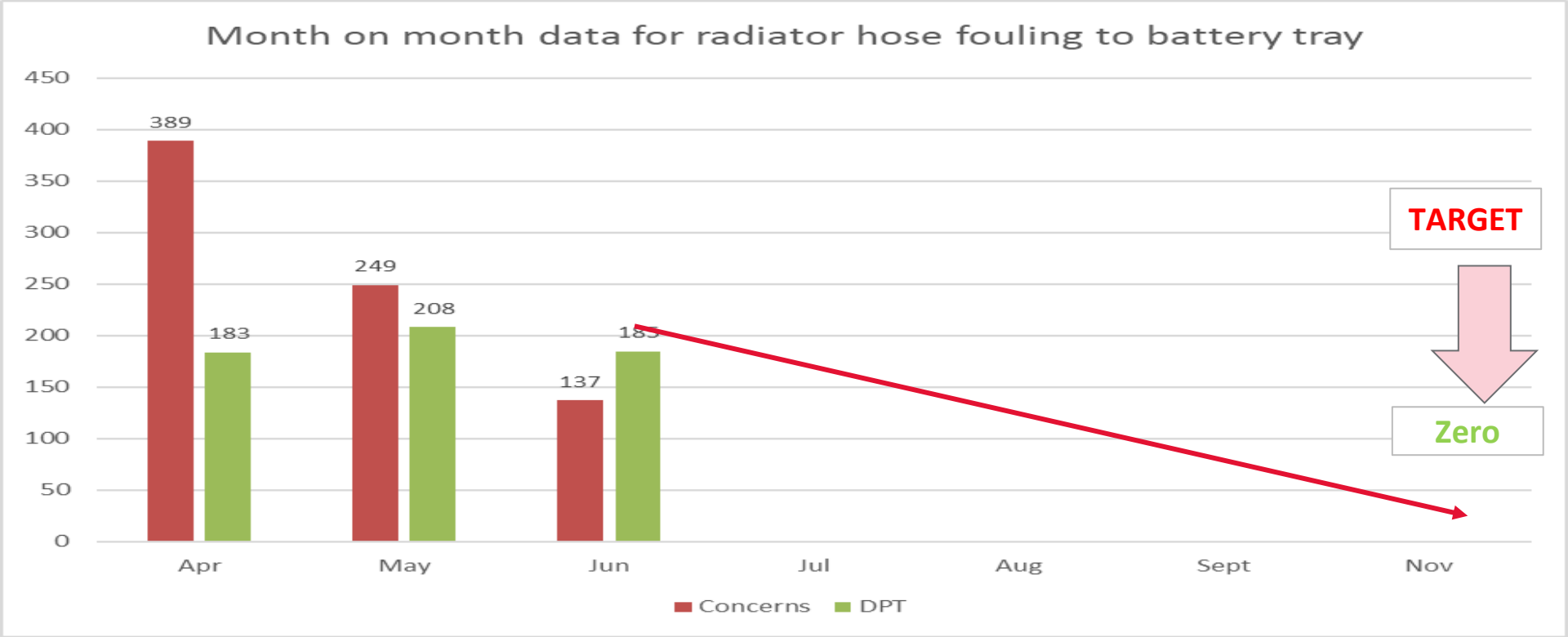
Battery tray



4. Select targets

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	775 cases reported based on data of last 03 months
Target :	**Concern elimination by end of Nov 2019**



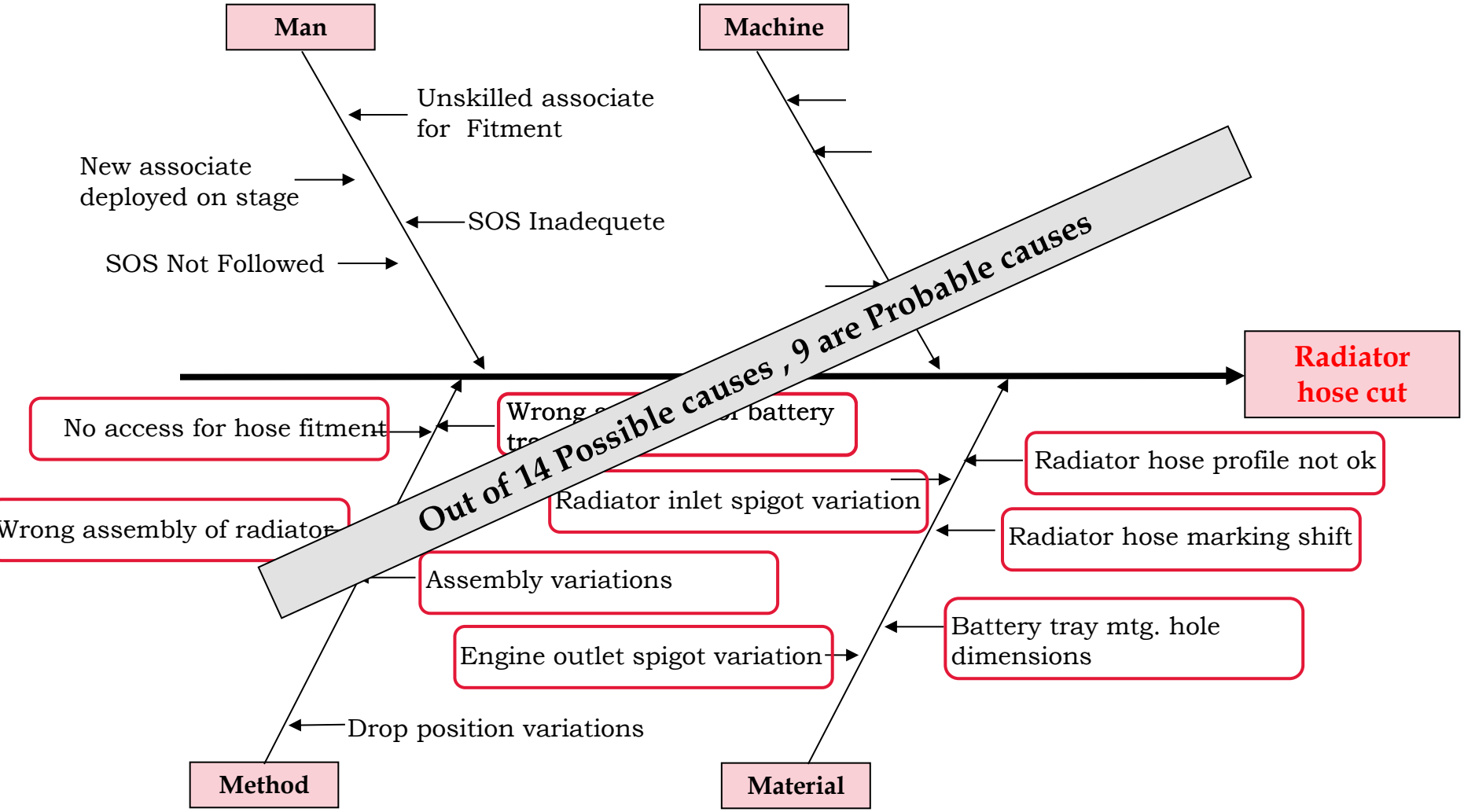
5. Analysis

List of Possible causes after brainstorming among team members

Sr. No	Possible Causes
1	New associate deployed at stage.
2	SOS Not Followed .
3	Unskilled associate for fitment.
4	No access for hose fitment
5	Wrong assembly of radiator hose
6	SOS inadequate
7	Wrong assembly of battery tray
8	Radiator hose profile not ok
9	Radiator hose marking position shift
10	Battery tray mtg. hole dimensions not ok
11	Assembly variation
12	Drop position Variation between body & chassis
13	Radiator inlet spigot variation
14	Engine outlet spigot variation

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr.	Possible Causes	Vikrant	Sandeep	Sayali	Bhaupatil	Mandar	Gokul	Total
1	New associate deployed at stage.	1	1	3	3	3	3	14
2	SOS Not Followed .	1	1	1	1	1	1	6
3	Unskilled associate for fitment.	1	1	3	1	1	1	8
4	No access for hose fitment	9	3	9	3	1	9	34
5	Wrong assembly of radiator hose	9	9	9	3	3	3	36
6	SOS inadequate	1	1	1	1	1	1	6
7	Wrong assembly of battery tray	3	3	9	3	9	9	36
8	Radiator hose profile not ok	3	9	9	3	3	9	36
9	Radiator hose marking position shift	9	9	9	3	1	3	34
10	Battery tray mtg. hole dimensions not ok	1	3	3	9	9	9	34
11	Assembly variation	1	3	3	1	1	1	11
12	Drop position Variation between body & chassis	9	3	1	1	9	9	32
13	Radiator inlet spigot variation	3	3	3	9	9	9	36
14	Engine outlet spigot variation	1	1	3	3	3	3	14

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 33.

Threshold limit - 30

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Possible Causes	Reasons
1	New associate deployed at stage.	All trained associate deployed on stage
2	SOS Not Followed .	SOS followed - Obsevation done by CFT
3	Unskilled associate for fitment.	All trained associate deployed on stage
6	SOS inadequate	Observation done by CFT found Ok
12	Drop position Variation between body & chassis	Observation done by CFT found Ok

TEST OF HYPOTHESIS

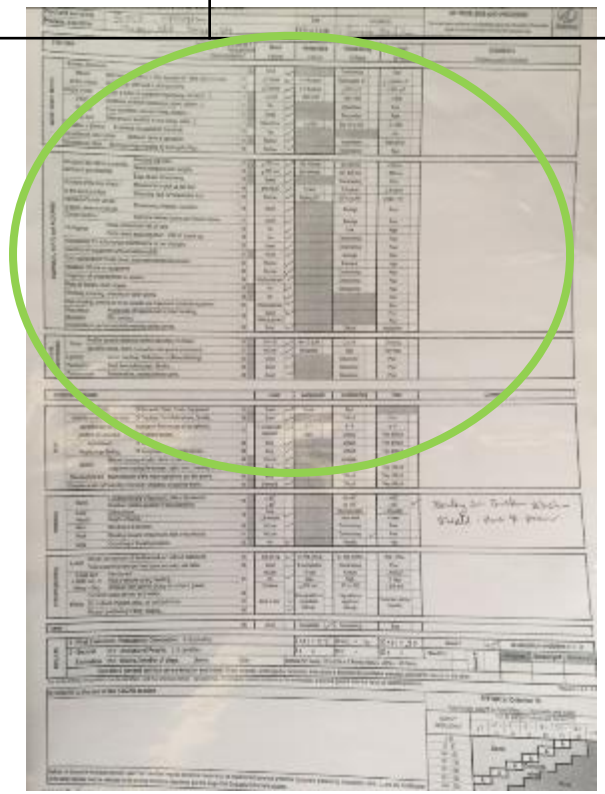
5.2 Confirm the probable cause:

Sr. no	Possible Causes
4	No access for hose fitment
5	Wrong assembly of radiator hose
7	Wrong assembly of battery tray
8	Radiator hose profile not ok
9	Radiator hose marking position shift
10	Battery tray mtg. hole dimensions not ok
11	Assembly variation
13	Radiator inlet spigot variation

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	No access for hose fitment	Ergonomic study of connection fitment	As per ergonomic study done- Though the operation is done inside engine compartment it is within acceptable limit	**Hypothesis invalid**



Observations :

- Access present for radiator hose fitment

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Wrong assembly of radiator hose	Visual inspection	Observation done on failed vehicles done. All hose fitted on marking of engine and radiator	**Hypothesis invalid**

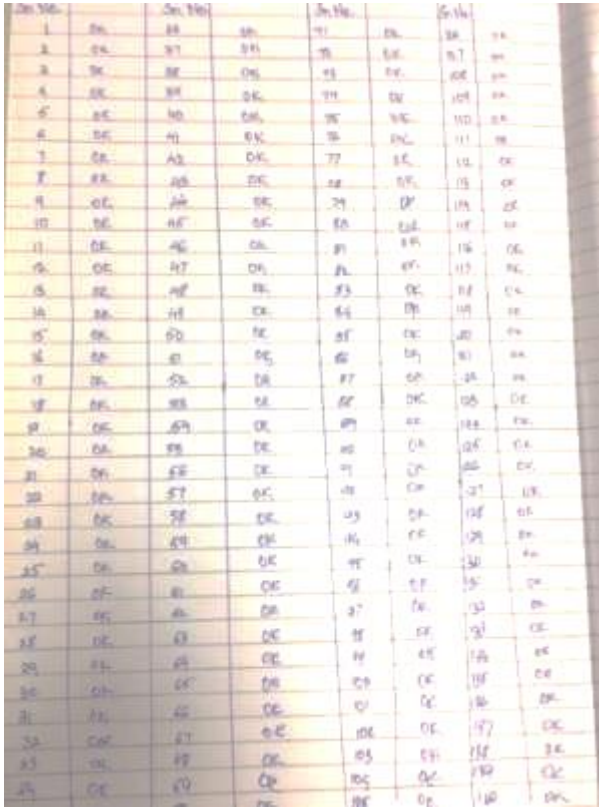
Observations :

- All assemblies found OK.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Wrong assembly of battery tray	Visual inspection	Observation done for 140 parts.	**Hypothesis invalid**



Observations :

- All assemblies found OK.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Radiator hose profile not ok	Using checking jig of supplier	Failed part and ok part fitted on jig	**Hypothesis valid**

Observations :

- Hose profile ok of both failed and OK part but minor difference at engine end as shown

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Radiator hose marking position shift	Using checking jig of supplier	Failed part and ok part fitted on jig	**Hypothesis invalid**

OBSERVATION OF NOT OK PART

Paint Marking OK

OBSERVATION OF OK PART

Paint Marking OK

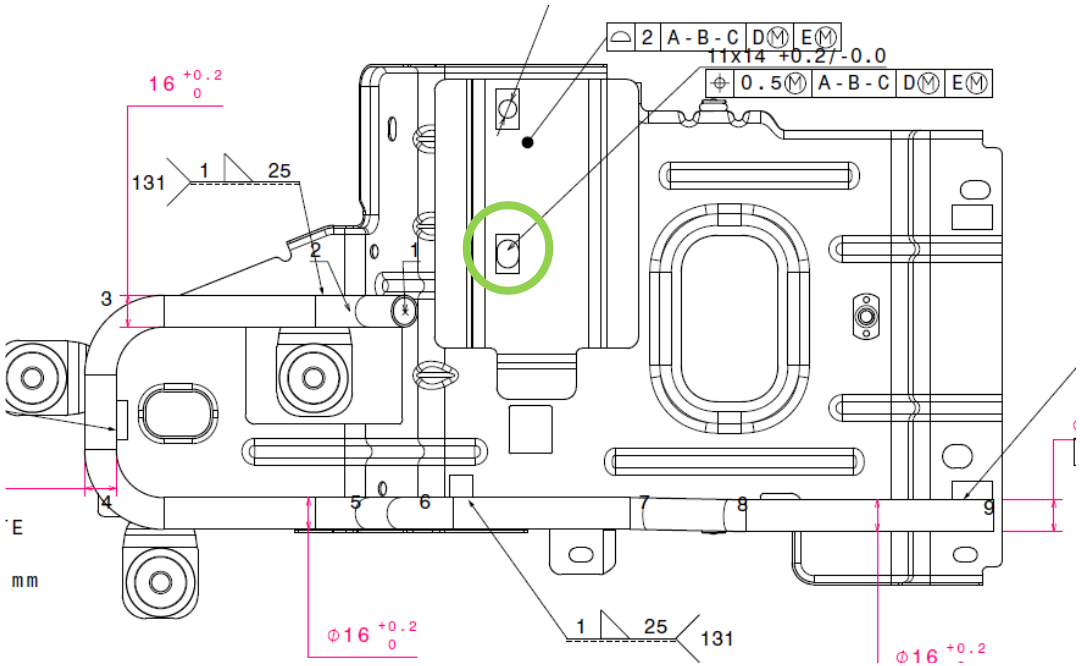
Observations :

- Marking position of failed part and Ok part matching.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	Battery tray mtg. hole dimensions not ok	Vernier	Observation done for 40 parts. All parts found Ok	**Hypothesis invalid**



Sr	11+2	14+2	Sr	11+2	14+2
1	11	14	21	11	14
2	11	14	22	12	15
3	12	15	23	13	16
4	13	16	24	12	15
5	12	15	25	12	15
6	10	13	26	13	16
7	9	12	27	12	15
8	10	13	28	11	14
9	11	14	29	11	14
10	12	15	30	11	14
11	11	14	31	12	15
12	9	12	32	13	16
13	10	13	33	12	15
14	11	14	34	13	16
15	11	14	35	11	14
16	12	15	36	11	14
17	13	16	37	12	15
18	12	15	38	13	16
19	10	13	39	12	15
20	11	14	40	11	14

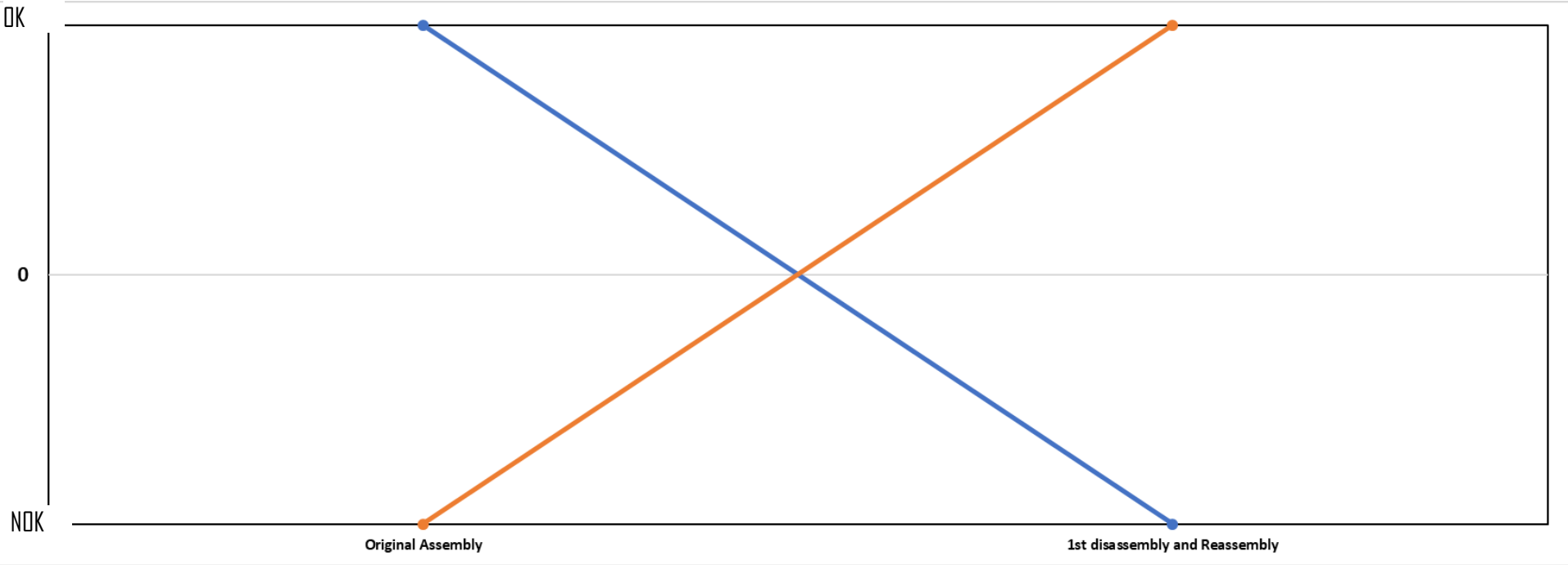
Observations :

- All parts found within tolerance

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Assembly variation	Assembly process validation done by CFT using DOE	OK vehicle and NOK vehicle change after remove refit	**Hypothesis valid**



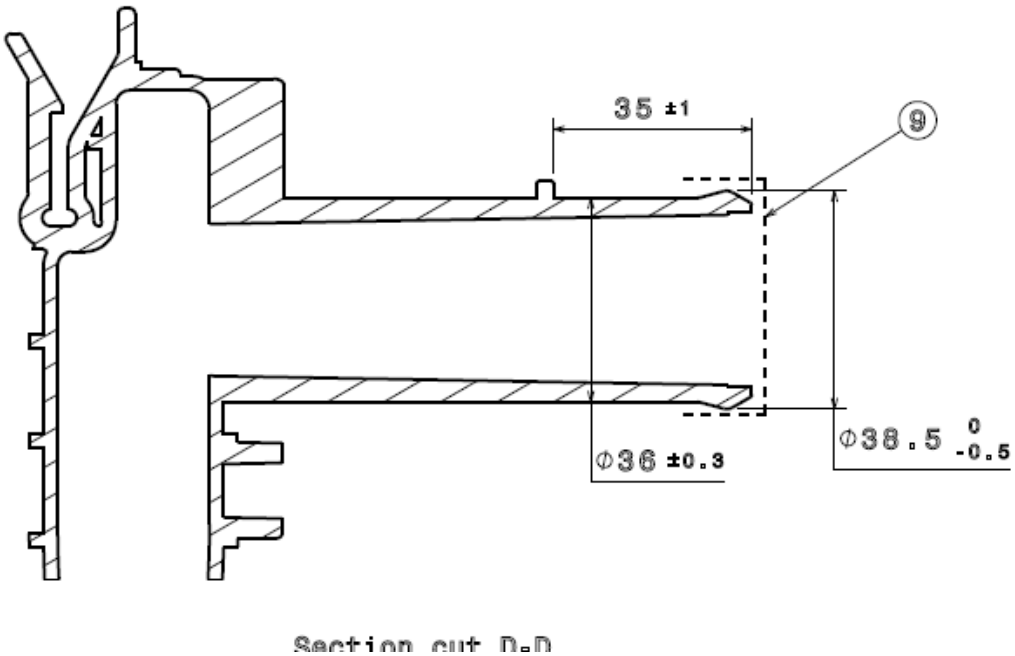
Observations :

- Assembly process causing fouling/gap less between battery tray and radiator hose

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	Radiator inlet spigot variation	Length checked using Vernier	Observation done for 40 parts. All found with in spec.	**Hypothesis invalid**



Sr	35±1	Sr	35±1
1	35	21	35
2	35	22	35
3	36	23	36
4	35	24	35
5	36	25	36
6	34	26	34
7	33	27	33
8	34	28	34
9	35	29	35
10	36	30	36
11	35	31	35
12	33	32	33
13	34	33	34
14	35	34	35
15	35	35	35
16	36	36	36
17	35	37	35
18	36	38	36
19	34	39	34
20	35	40	35

Observations :

- All length found with spec 35±1mm

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. no	Possible Causes	Testing Method	Conclusion
1	No access for hose fitment	Visual	Hypothesis Invalid
2	Wrong assembly of radiator hose	Visual	Hypothesis Invalid
3	Wrong assembly of battery tray	Visual	Hypothesis Invalid
4	Radiator hose profile not ok	Checking Jig	Hypothesis Valid
5	Radiator hose marking position shift	Checking Jig	Hypothesis Invalid
6	Battery tray mtg. hole dimensions not ok	Vernier	Hypothesis Invalid
7	Assembly variation	DOE	Hypothesis Valid
8	Radiator inlet spigot variation	Vernier	Hypothesis Invalid

WHY - WHY Analysis

5.3 Identify the root causes:

Sr. No.	Probable cause	WHY?	WHY?	WHY?	Action
1	Radiator hose profile not ok	End caps not mounted properly with locking arrangement..	Bottom end cap go down by 4 to 5mm	Bottom end cap lock with nut and bolt arrangement only.	**Bottom end capping lock to be changed.**
2	Assembly variation	Associates pushes hose towards bracket	To fit radiator hose on engine end	As per process	**Need to change process**

6. Implement the corrective measures

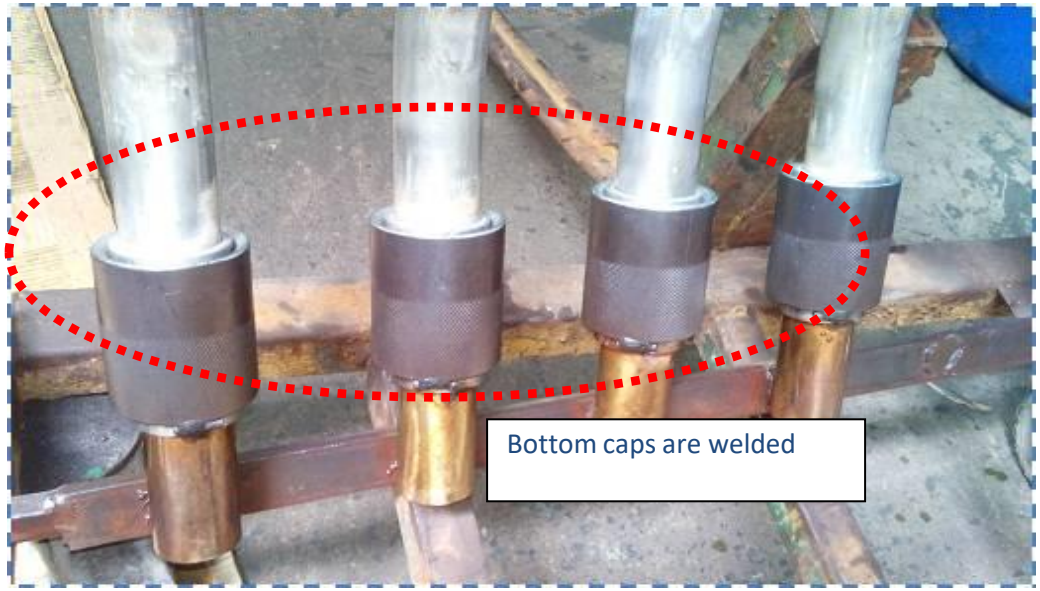
Possible Remedies Identified

Sr. No.	Root cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Radiator hose profile not ok	**Bottom end capping lock to be changed.**	Medium	Low	Yes	Low	**Accepted**
2	Assembly variation	**Need to change process**	Low	Low	Yes	Low	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Radiator hose profile not ok	**Bottom end caps welded**	24.07.19	Mr. Abhijit Puranik	Closed



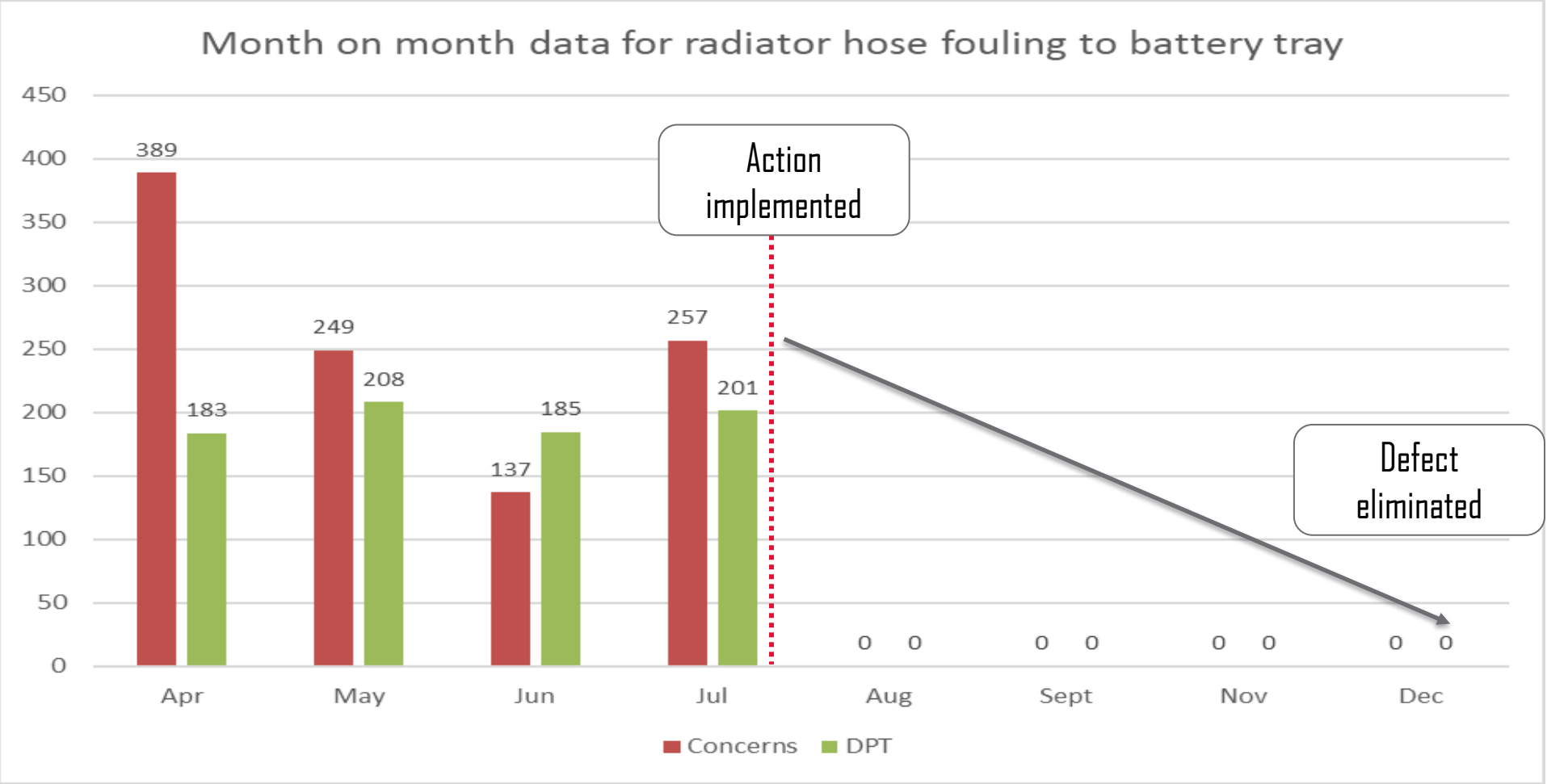
6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	Assembly variations	**Radiator hose fitment at engine end shifted to chassis line**	01.08.19	Mr. Vikrant Ghule	Closed

7. Check & confirm the results & effects

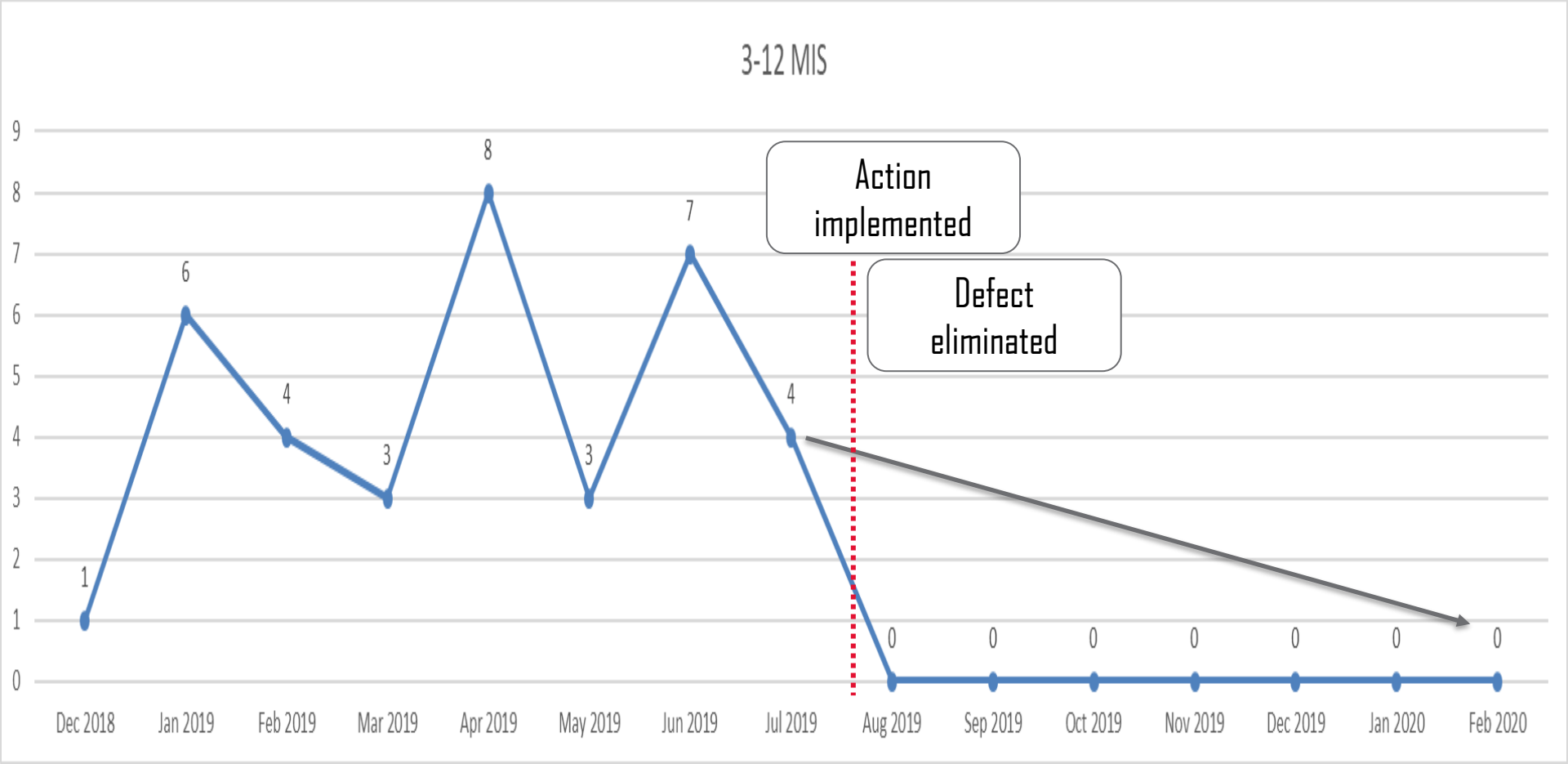
7.1 Verify Results:



Inference – As per data of 3 months concern is eliminated

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – As per data of 3 months concern is eliminated

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Plan	Actual date	Responsibility	Status
1	SCAP	24.07.19	Abhijit Puranik	Closed
2	Checksheet	01.08.19	Vikrant Ghule	Closed
3	SOS	01.08.19	Vikrant Ghule	Closed
4	PFMEA	24.07.19	Abhijit Puranik	Closed

Check & confirm the results & effects (Cont……)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of radiator hose cut in Marazzo model
2) Rework cost is reduced by approx. Rs.96774.19/Annum.

In tangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learned
3) Associate Morale Increased
4) Improved Customer Satisfaction.
5)Less rework fatigue

RE: Rework Cost estimation

DHANOKAR RAJESH
To GHULE VIKRANT

Ok Approved

From: GHULE VIKRANT
Sent: Thursday, February 27, 2020 1:35 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: Rework Cost estimation

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	10
Total Required Rework time per day	Min	100
Effective working time	Min	465
Operator required /day	nos	0.22
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	322.58
Avg no. of working days	days	25
Operator cost /month	Rs	8064.52
Operator Cost/year	**Rs**	**96774.19**

Total Rework cost	Rs / Year	96774

8. Standardization

8.1 Updation of documents :-

S N	Plan	Actual date	Responsibility	Status
1	SCAP	24.07.19	Abhijit Puranik	Closed
2	Check sheet	01.08.19	Vikrant Ghule	Closed
3	SOS	01.08.19	Vikrant Ghule	Closed
4	PFMEA	24.07.19	Abhijit Puranik	Closed

8.2 Training:

S N	Training Identified	Actual date	Responsibility	Status
1.	Stage & buy off rework associates awareness training conducted by explaining the changes done in breather pipe	01.09.18	Sandeep Kate	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Plan	Actual date	Responsibility	Status
1	SCAP	24.07.19	Abhijit Puranik	Closed

TO BE FILLED BY M&M

M & M Ltd. Automotive Sector | **SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)** | **PU Name : J1 ETA Sr. No: Date: 12.06.2019**

Supplier : IMPERIAL AUTO INDUSTRIES LTD. | Supplier Code: DI005
Part Description: Radiator hose | Part No.:0304DAB02060N
When Observed (Date / Period) - 30.04.2019 | GRIInvoice No.: --

A PROBLEM STATEMENT : Should Answer | ISSUE REPORTED : REPEAT / 1'ST TIME ☑

What to the problem? Length variation
Who Is affected - Customer
Where Location/ Issue
When Issue.
Why. will affect performance
How much Qty. Data

Problem Statement -Radiator Hose fouling

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.

Radiator Hose fouling with Battery bracket.

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)
5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Not detected during final inspection.	Profile and length checked at final inspection on samples/lot	Manufactured in capping mandrel as such the possibility of variation is remote.		

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:
C1:inhouse stock verified 85 nos checked and found ok.

C2: Segregation Status:

Location	Ok Qty.	Not Ok qty.	Conformity mark (show location by Picture or photo)
At Customer end	200	0	
IAI Warehouse			
Tansit			
At IAI	85	1	

D D1) ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine
Is the specified M/C used ? ☑ YES ☐ NO
Is M/C in good condition ? ☑ YES ☐ NO
Is it a substitute M/C ? ☐ YES ☑ NO
Are M/C parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO
***

Man
Is operator trained ? ☐ YES ☑ NO
Is operator is under training ? ☑ YES ☐ NO
Is operator temporary ? ☐ YES ☑ NO
Is it due to Bad Rework ? ☑ YES ☑ NO

Operator Under Training

Measurement
Are instruments / gauges calibrated ? ☑ YES ☐ NO
Are Control Methods Capable to detect Non conformity? ☐ YES ☑ NO
Are specified Control Methods used ☐ YES ☐ NO

Sampling Inspection at Final.

Length Variation

Method
Are Control Plan/SOP available near M/C? ☑ YES ☐ NO
Are Control Plan respected ? ☐ YES ☑ NO
Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO
***

Material / Tooling
Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
Is the Raw Matl./Tooling Supplier change ? ☐ YES ☐ NO
Is the heat treatment method change ? ☐ YES ☐ NO
Is Tool surface condition ok ? ☐ YES ☐ NO

Environment
Is process needs environmental control? ☐ YES ☑ NO
Is it implemented ? ☐ YES ☐ NO
Does it conform ? ☐ YES ☐ NO
***

*** Space provided for any other cause

1.End caps not mounted properly by the operator.
2.Not Detected at Final Inspection due to sampling Inspection.

D3) 5 WHY ANALYSIS (ANALYSIS FOR PRODUCING NOT OK PART)

	WHY	WHY	WHY	WHY	WHY
1	Length Variation	End caps not mounted properly with locking arrangement.	Bottom end cap go down by 4 to 5mm	Bottom end cap lock with nut and bolt arrangement only.	
2					
3					
4					

E CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	Bottom end cap lock with nut and bolt arrangement.	End Capping Fixed with welding and nut/bolt from bottom side to avoid movement from bottom side.	SANJIV	17.06.2019	Done
2	Profile and Length checked on sample basis in the fixture.	100% Inspection will be carried out.	PRADEEP	17.06.2019	Done
		End Capping Fixed from bottom side to avoid movement from bottom side.	Shasi Mishra	18.06.2019	Done

F ACTION TAKEN TO PREVENT RECURRENCE
On job training will be organized in concerning area.
Supervisor concerned will have special attention when additional manpower on job.
100% Inspection will be done at final inspection for profile and length.

G CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

All Radiator & Heater with capping mandrels.

H Following quality documents changed due to implemented action plan : (mention doc. no.)

Process Flow Chart ☐ | Gauge Drawing ☐
PFMEA ☑ | Fixture Drawing ☐
Control Plan ☑
OPL / SOP ☐

I EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem ☐ YES ☐ NO
Are actions planed completed / implemented ☐ YES ☐ NO
Is problem repeated in lots received after SCAP issue Date ☐ YES ☐ NO

FOR MAHINDRA & MAHINDRA (A.S.) | SUPPLIER'S REPRESENTATIVE

Signature / Date | Signature / Date

MQS/SCAP/ F/01 | Rev.02 | 06/25/07

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Plan	Actual date	Responsibility	Status
2	Checksheet	01.08.19	Vikrant Ghule	Closed

15		Radiator hose fitment on raditor	Coplete inserted Matche marking matched	V1	Visual ,Marking	Field		Relative Marking
16		Radiator hose fitment on engine	Not fouls to Battery Tray	V1	Visual ,Marking	Field		Relative Marking
Rev No.		Date	Revision Details			Reason for Revision		

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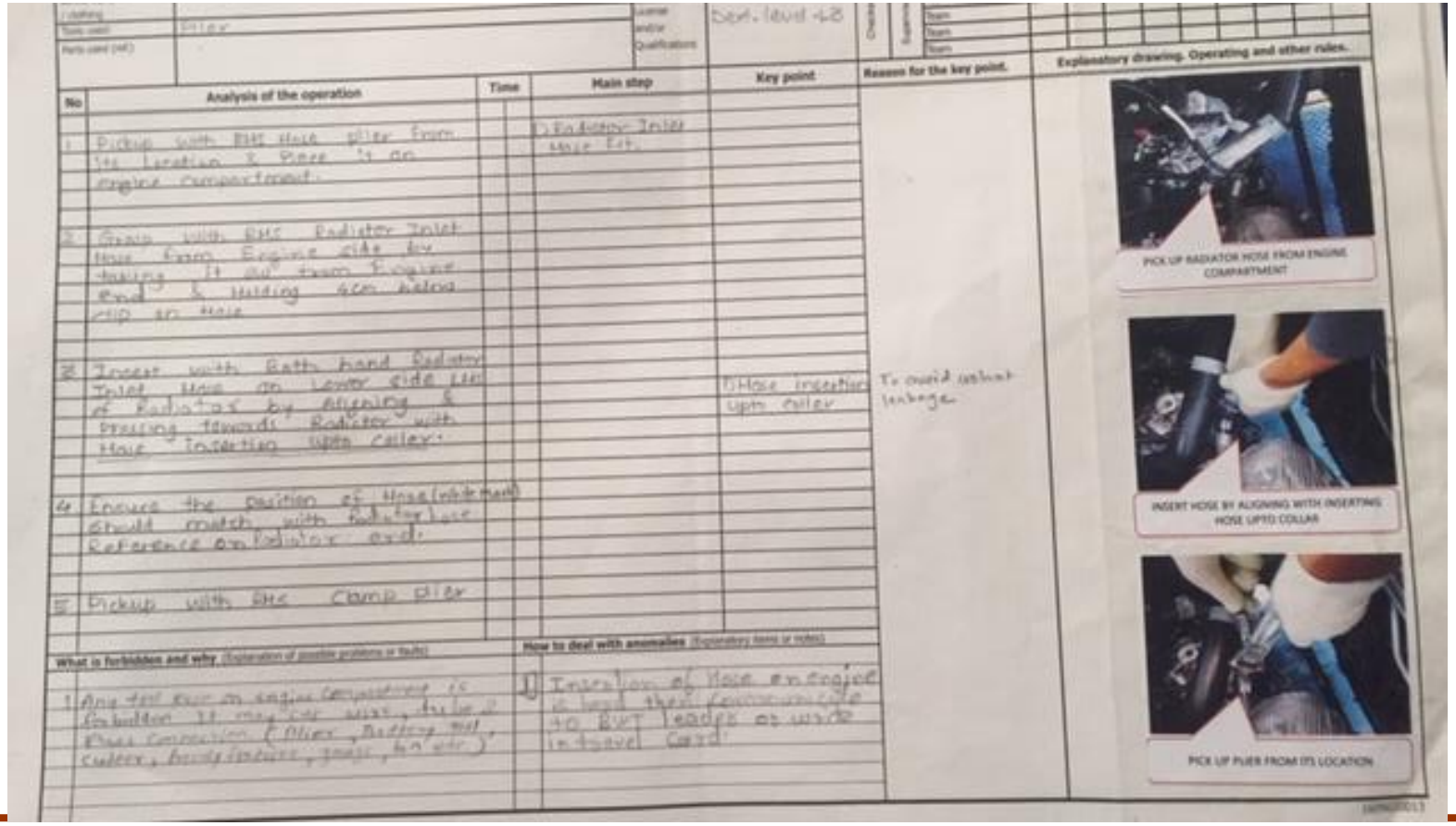
8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Plan	Actual date	Responsibility	Status
3	SOS	01.08.19	Vikrant Ghule	Closed



Plier

No	Analysis of the operation	Time	Main step	Key point	Reason for the key point
1	Pickup with RHS Hose plier from its location & Place it on engine compartment.		1) Radiator Inlet Hose Fit.		
2	Grasp with RHS Radiator Inlet Hose from Engine side by taking it out from Engine end & holding 4cm extra grip on Hose				
3	Insert with Both hand Radiator Inlet Hose on Lower side pipe of Radiator by aligning & pressing towards Radiator with Hose Insertion upto collar.			1) Hose insertion upto collar	To avoid water leakage.
4	Ensure the position of Hose (white mark) should match with Radiator hose Reference on Radiator end.				
5	Pickup with RHS Clamp plier				

What is forbidden and why (Explanation of possible problems or faults)

1) Any tool kept on engine compartment is forbidden. It may cut wire, tube & Hose connection (Plier, Battery nut, cutter, body rubber, grease, tyre etc)

How to deal with anomalies (Regulatory items or rules)

1) Insertion of Hose on engine side head that connection side to Burr leaders or wire in travel Card.

Explanatory drawing. Operating and other rules.

PICK UP RADIATOR HOSE FROM ENGINE COMPARTMENT

INSERT HOSE BY ALIGNING WITH INSERTING HOSE UPTO COLLAR

PICK UP PLIER FROM ITS LOCATION

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Plan	Actual date	Responsibility	Status
4	PFMEA	24.07.19	Abhijit Puranik	Closed

iAi

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS(PROCESS FMEA)

FMEA Number: FE/XXV/DT-RS/01
Prepared By:- Pardeep Kumar
FMEA Date(Orig.) : 01.01.2012

Item: Rubber Hose - MASTER PFI Process Responsibility: Production
Model Year Key Date: 17.05.2019 FMEA Revision :- 122
Core T Mr. B.Verma, Amit Sharma, Sonu Raina, Ashok Dabas, Rajesh Gupta, Rajkumar, R. Panday, K.C. Gola, S.Bhandari, Pardeep

Proce / nction	Sub Process	Requirements	Potential Failure Mode	Potential Effect(s) of Failure N- Next Operation S- Subsequent operation C-Customer End E-End User	Severity	Classification	Potential Cause(s) of Failure	Current Process Controls Prevention	Occurrence	Current Process Controls Detection	Detection	S X O	RPN	Recommended Action	Responsibility & Target Completion Date	Action Result: Actions Taken & Effective Date	Severity	Occurrence	Detection	RPN
		Length as per spec	Length Over	N-Nil (1) S-Fitment Problem (7) C-Fitment Problem (7) E-Nil (1)			Bottom Cap not set in mandrel	Work Instruction Skill Matrix Lock to control the cap position provided in Mandrels.	2	Visual check	7	14	98	*End Capping Fixed with welding and nut/bolt from bottom side to avoid movement from bottom side.*	*SANJEEV 17.05.2019*	*IMPLEMENTED 17.05.2019*				

9. Summarize & Plan future actions

9.1 Conclusion:

Elimination of concern of radiator hose cut from Marazzo model

9.2 Horizontal Deployment:

All Marazzo models having similar design. Routing changed in BS6

9.3 Lessons Learnt:

Things Gone Wrong :

- Initially more emphasis went on design.
- While designing any part critical clearance due fitment variations need to be checked.

Things Gone Right :

- Process improvement done on fast track.
- Simple solutions giving big benefits

Thank you

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